Simulated motor - mechanical troubleshooting

FICTIONAL TRAINING MANUAL - NOT A MANUFACTURER DOCUMENT

Applicability: Prototype motor MTR-001, asset simulation-01, Simulator.html and the Python server model. These fictional instructions apply only to virtual controls. They are not approved physical-equipment limits, maintenance specifications, or evidence of a real fault.

READING THE SIGNAL

Channels: temperature in degC; motor current in A; vibration RMS and acceleration X/Y/Z in g; gyroscope X/Y/Z in deg/s. Published rpm is the selected setpoint. Effective rotation is reduced internally for Overload and Stall. Acceleration RMS in g must not be relabeled as velocity in mm/s.

The simulator publishes non-retained MQTT sensor windows. Each window includes min, mean, max and sample count. Waveforms contain the faster model samples; slow dashboard summaries do not prove a harmonic or spectral frequency.

DETECTION AND ILLUSTRATIVE LABELS

Each channel has its own rolling z-score against previous valid values. The default detector warms up over five windows, uses up to twenty baseline values, and flags an absolute score of at least three. A standard-deviation floor prevents division by zero. A statistical deviation does not identify its physical cause.

Illustrative model labels use vibration above 0.15 g or temperature above 60 degC or relative current above 1.35 for WARNING, and vibration above 0.35 g or temperature above 75 degC or relative current above 1.9 for FAULT. Relative current uses the selected-speed healthy-current model. These demonstration labels are independent of the rolling detector and are not approved operating limits.

Normal operation and baseline drift

EXPECTED SIMULATED SIGNATURE

Healthy virtual motor: small shaft-frequency vibration and residual sensor noise. At a selected speed of 1500 rpm, modeled current settles near 1.6 A and the thermal target is 37 degC. Initial temperature is 28 degC and approaches the target gradually. A warming motor can trigger a statistical temperature deviation even with Normal selected.

DISTINGUISHING EVIDENCE

A rolling z-score describes change relative to recent values. It is not a confirmed fault, an ISO vibration classification, or an approved machine limit. Constant high readings can become statistically normal as the baseline adapts.

VIRTUAL TROUBLESHOOTING CHECKS

Select Normal in the simulator and allow the temperature trend to settle.

Compare the selected shaft speed with the recorded speed before interpreting current.

Review the previous valid windows before interpreting a baseline deviation.

RECOVERY AND LIMITATIONS

Compare repeat windows at the same selected speed and severity. Record the warm-up state and retain the initial event as a historical deviation.

Rotor imbalance

EXPECTED SIMULATED SIGNATURE

Rotor imbalance increases acceleration X and Y at the shaft frequency, also called the once-per-revolution or 1x component. The imbalance contribution grows with severity and the square of effective speed. Vibration RMS rises while mean current changes only slightly. Temperature target has a small additional rise.

DISTINGUISHING EVIDENCE

Dominant radial once-per-revolution motion with relatively modest current change supports an imbalance hypothesis. It differs from the twice-per-revolution axial motion used by Misalignment and the sharp impacts used by Bearing fault.

VIRTUAL TROUBLESHOOTING CHECKS

Compare acceleration X and acceleration Y with the Normal waveform at the same selected speed.

Reduce the simulated speed and compare the change in vibration RMS.

Return the fault selector to Normal and verify that radial vibration declines.

RECOVERY AND LIMITATIONS

A declining radial waveform after removing the virtual imbalance supports the model experiment. The simulation does not prove a real rotor needs balancing.

Shaft and coupling misalignment

EXPECTED SIMULATED SIGNATURE

Misalignment produces a twice-per-revolution or 2x harmonic in acceleration X and Z, an additional Y contribution, heating, and an approximate severity-dependent current rise. Axial acceleration Z must be interpreted relative to its gravity offset near 1 g, not as an absolute vibration magnitude.

DISTINGUISHING EVIDENCE

Axial motion and a strong twice-per-revolution component distinguish misalignment from the radial imbalance experiment. Gyroscope channels respond to the modeled vibration but are not an independent confirmed alignment measurement.

VIRTUAL TROUBLESHOOTING CHECKS

Compare acceleration Z around its gravity offset with the Normal waveform.

Compare the axial waveform and current trend while changing the simulated misalignment severity.

Select Normal and verify that the extra harmonic contribution subsides.

RECOVERY AND LIMITATIONS

Record the changed axial waveform and temperature trend. Temperature may remain elevated while the first-order model cools.

Bearing fault and friction impulses

EXPECTED SIMULATED SIGNATURE

Bearing fault generates periodic acceleration impulses with a short ring-down, at approximately 3.57 times effective shaft frequency. Impacts enter X, Y, and Z with different amplitudes. Friction increases the thermal target and mean current; current also contains impact-related ripple.

DISTINGUISHING EVIDENCE

Periodic sharp impacts and decaying high-frequency oscillations distinguish this case from the smoother imbalance waveform and the irregular Loose mount knocks. The simulator does not identify an actual bearing part number or prove a raceway defect.

VIRTUAL TROUBLESHOOTING CHECKS

Inspect the simulated acceleration waveform for repeated impacts and decaying oscillations.

Compare current and temperature trends alongside the vibration RMS deviation.

Reduce bearing-fault severity and compare the impact amplitudes before returning to Normal.

RECOVERY AND LIMITATIONS

Verify declining impacts across repeat virtual windows. Preserve the evidence and avoid interpreting a single sampled axis mean as the complete impact waveform.

Loose mounting and irregular knocks

EXPECTED SIMULATED SIGNATURE

Loose mount creates random knocks with ring-down and a half-shaft-frequency or 0.5x subharmonic. Effects appear mainly in X and Z acceleration, with small impact-related current spikes. Randomness means adjacent windows can differ substantially at identical controls.

DISTINGUISHING EVIDENCE

Irregular timing and a half-frequency rattle distinguish loose mounting from the periodic Bearing fault experiment. A current spike can accompany a mechanical knock without establishing an electrical short.

VIRTUAL TROUBLESHOOTING CHECKS

Compare several simulated windows for irregular knocks instead of relying on a single point.

Compare acceleration X and Z with the current trend during the loose-mount experiment.

Select Normal and verify that irregular knock activity subsides over repeat windows.

RECOVERY AND LIMITATIONS

Record the removed virtual fault and compare repeat windows. This guide supplies no physical mounting torque or fastener specification.

ACQUISITION AND STATISTICAL EDGE CASES

MISSING OR STALE READINGS

Check whether the server, MQTT broker and publisher are connected. A paused model, stopped publisher or disconnected broker can leave old points visible. An old point is history and must not be reported as a fresh normal observation. Resume the publisher and wait for a new valid timestamp.

SENSOR ERROR OR DISCONNECTED SENSOR

A sensor error has null features and zero samples. Missing data is not zero current or a cold temperature. Check the sensor status, sample count and unit before interpreting an apparent deviation.

DUPLICATE, OVERLAPPING OR REJECTED WINDOWS

QoS redelivery can repeat a sequence. The backend ignores duplicate or out-of-order identities and rejects overlapping, stale, future-dated or invalid windows. A missing graph point can indicate rejected transport data rather than a mechanical fault. Compare the publisher and backend clocks and review the received sequence.

STARTUP, CONTROL CHANGES AND BASELINE ADAPTATION

Warm-up, RPM changes, a changed fault severity, and the cool-down reset can all create deviations. Sustained abnormal values can enter the rolling baseline and later receive a NORMAL label. Record the selected mode and control changes alongside every anomaly.

STOPPING AND RESTARTING

Setting selected speed to zero drives modeled current toward zero and removes shaft-driven vibration. This intentional shutdown can differ from the previous rolling baseline. A restarted backend rebuilds its history and baseline. Compare like operating conditions and do not invent measurements across a gap.